

Each of the IoT devices will require, at the minimum, a micro-controller to add intelligence to the device, one or more sensors for data collection, one or more chips for connectivity and data transmission, and a memory component. Connected devices that transmit information across relevant networks rely on innovations from semiconductor players—highly-integrated microchip designs, for instance, and very low-power functions in certain applications.

Semiconductor companies that can effectively deliver these and other innovations to original-equipment manufacturers, original-device manufacturers and others that are building IoT products and applications will play an important role in the development of the market. That market, in turn, may represent a significant growth opportunity for semiconductor players and their customers.

Shift in technology

Wide adoption of the IoT may need a new class of systems on chip (SoC) based devices, with provision for optimal power and connectivity as well as sensor integration. The first generation of such chips is already on the way, although it will probably be a few generations before chips can deliver all the required functionality.

Additionally, a new array of sensors based on micro-electromechanical systems (MEMS) technology are rapidly being developed to enable IoT applications beyond motion and image sensing to include those that measure humidity, altitude, food calorie composition and human health indicators. This is a result of the increasing variety of consumer IoT applications such as wearables and clustered systems.

Connectivity being a key capability required in IoT applications, multiple communications standards and protocols are being

Survey participants

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- Dr Amarjeet Singh, chief technology officer, Zenatix Solutions Pvt Ltd
- Anand T., managing director, KNEWRON Technologies Pvt Ltd
- Ashish Gulati, country head, India, Telit
- Chandramouli Srinivasan, chief executive officer, Mobodexter
- Dileep Jain, sales head, Rajguru Electronics
- Gopinath Y., chief business officer, ConnectM
- Hemant Patil, head-CoE - manufacturing IT, Tata Technologies
- Jayakrishnan K.B., co-founder, Wisys Technologies
- Krupa Shukla, chief executive officer, Yups Tech Solutions Pvt Ltd
- Ming Yu, business development manager, Moxa India Industrial Networking
- M. Vinod Kumar, staff engineer, Tejas Networks Ltd
- Niraj Tyagi, head-CoE - manufacturing, Tata Technologies
- Payal Sharma, marketing manager, Invendis Technologies India Pvt Ltd
- P. Chow Reddy, managing director, Interleaved Technologies
- Rakesh Varun R., chief strategy officer, Hyperthink Systems
- Shachindra, chief executive officer, NeerAssure
- Syam Madanapalli, IEEE senior member, executive member of IEEE Bengaluru chapter
- Sreowsi Sinha, co-founder and director - marketing, Frugal Labs Tech Solutions Pvt Ltd
- Dr Srinivas Mandavilli, country manager, Imagination Technologies
- Srinivasa Moorthy, chief executive officer, Andhra Pradesh Electronics & IT Agency
- Sujata Tilak, managing director, Ascent Intellimination Pvt Ltd

supported by consumer electronics and industrial devices like low power and personal area wireless mesh networks, which are primarily suited for sensors.

IoT applications usually require a single chip at an acceptable form factor and low power consumption for battery-operated devices. This requirement is a factor for growth in SoCs with embedded cores, graphical processing units and integrated wireless connectivity in a single package.

Moreover, progress in these technologies has allowed the development of IoT devices such as sensors that have computing, storage and network capabilities built into extremely small form factors with low-energy requirements.

On the service front, one emerging requirement will be asynchronous data transmission by the device. This will lower the cost of the device and the entire infrastructure.

This will also need security in order to prevent tampering with the data transmitted by the device. And reduction in ISO-OSI stack will help; working closer to the PHY layer will

be better suited in terms of cost, power, carbon dioxide emissions and so on.

On the flip side

Like any other innovative concept, the growth of the IoT also depends on several factors. We asked the survey participants to suggest possible challenges that could derail the growth of the IoT in the country. Here is a collation of their opinions:

- Lack of awareness and limited availability of real products and services
- High cost of connectivity
- Inadequate security and privacy protection for user data
- Lack of consistent standards
- Technological limitations that affect the IoT's functionality
- Discontinued and fragmented IoT value chain
- Lack of cost-effective solutions

However, the hope is that these problems are not insurmountable, particularly if the electronics industry players are willing to take an active role in solving these. **EFY**